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Phenylalanine Interacts with Oleic Acid-Based Vesicle Membrane. Understanding the Molecular Role of Fibril–Vesicle Interaction under the Context of Phenylketonuria

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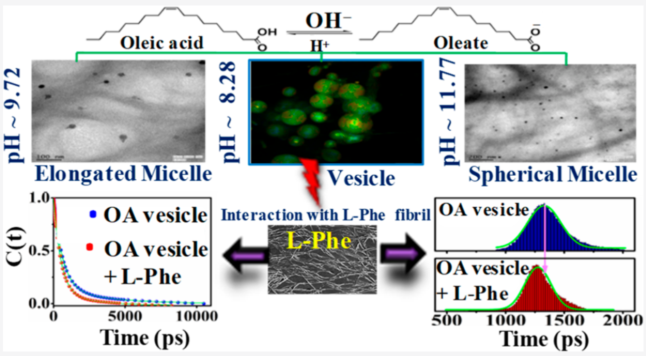
ABSTRACT: In the present contribution, on the basis of a spectroscopic and microscopic investigation, the characterization and photophysics of various assemblies of oleic acid/oleate solution at three pH values, namely, 8.28, 9.72, and 11.77, were explored. The variation in the dynamic response of aqua molecules in and around the assemblies has been interrogated by a picoseconds solvation dynamics experiment using a time-correlated single-photon counting setup employing coumarin-153 as a probe. On the one hand, the time-resolved fluorescence anisotropy measurement along with the fluorescence correlation spectroscopy experiment was executed to extract information regarding the comparison of the extent of the internal restricted confinement of these assemblies. On the other hand, an effort to investigate the cross-interaction between the self-assembled architectures of L-phenylalanine (L-Phe), responsible for phenylketonuria (PKU) disorder, and the oleic acid at the vesicle-forming pH established that the L-Phe fibrillar morphologies strongly alter the dynamic properties of the vesicle membrane formed by the oleic acid. Specifically, the interaction of the L-Phe assemblies with the oleic acid vesicle membrane is found to introduce the flexibility of the vesicle membrane and alter the hydration properties of the membrane. To track the fibril-induced alterations of the oleic acid vesicle properties, various spectroscopic and microscopic investigations were performed. The mutual reconciliation of the experimental outputs, therefore, portrays the state of the art, which accounts for the fibril-induced alterations of the properties of the oleic acid vesicle membrane, the mimicking setup of the cellular membrane, thereby informing us that alterations of such a property of the membrane should be taken into active consideration during the rational development of therapeutic modulators against disorders like PKU.



1. INTRODUCTION

The essential amino acid phenylalanine (Phe), under certain physiological abnormal functionalities, like phenylketonuria (PKU), forms the basis of the inborn error of metabolism, leading to the progression of neuropsychological and psychosocial abnormalities.[1] The molecular pathophysiology of PKU has been reported to be associated with the lack of appropriate functioning of the enzyme phenylalanine hydroxylase (PAH), which in turn results in the deposition of unprocessed Phe in various parts of the body in an untreated patient.[2,3] Specifically, it deposits in the forms of fibrillar morphology having the amyloid-like etiology of a toxic nature, which is a signature of PKU as well as Alzheimer's and related neurological disorders.[4−6] Such a formation of the fibrillar morphology of Phe under a physiological in vitro condition is well-established.[4,7] The formation of the toxic fibrillar aggregates have gotten recognition as a marker of these neurological disorders, like PKU;[4] however, extensive efforts from the researchers in this active field are still engaged to investigate the mechanism by virtue of which such fibrillar assemblies impact the cells under the PKU state.[8,9] One generic approach toward such an investigation of the effect of the Phe fibrils on the cellular environment is to set up the physiologically mimicked environment by preparing the lipid

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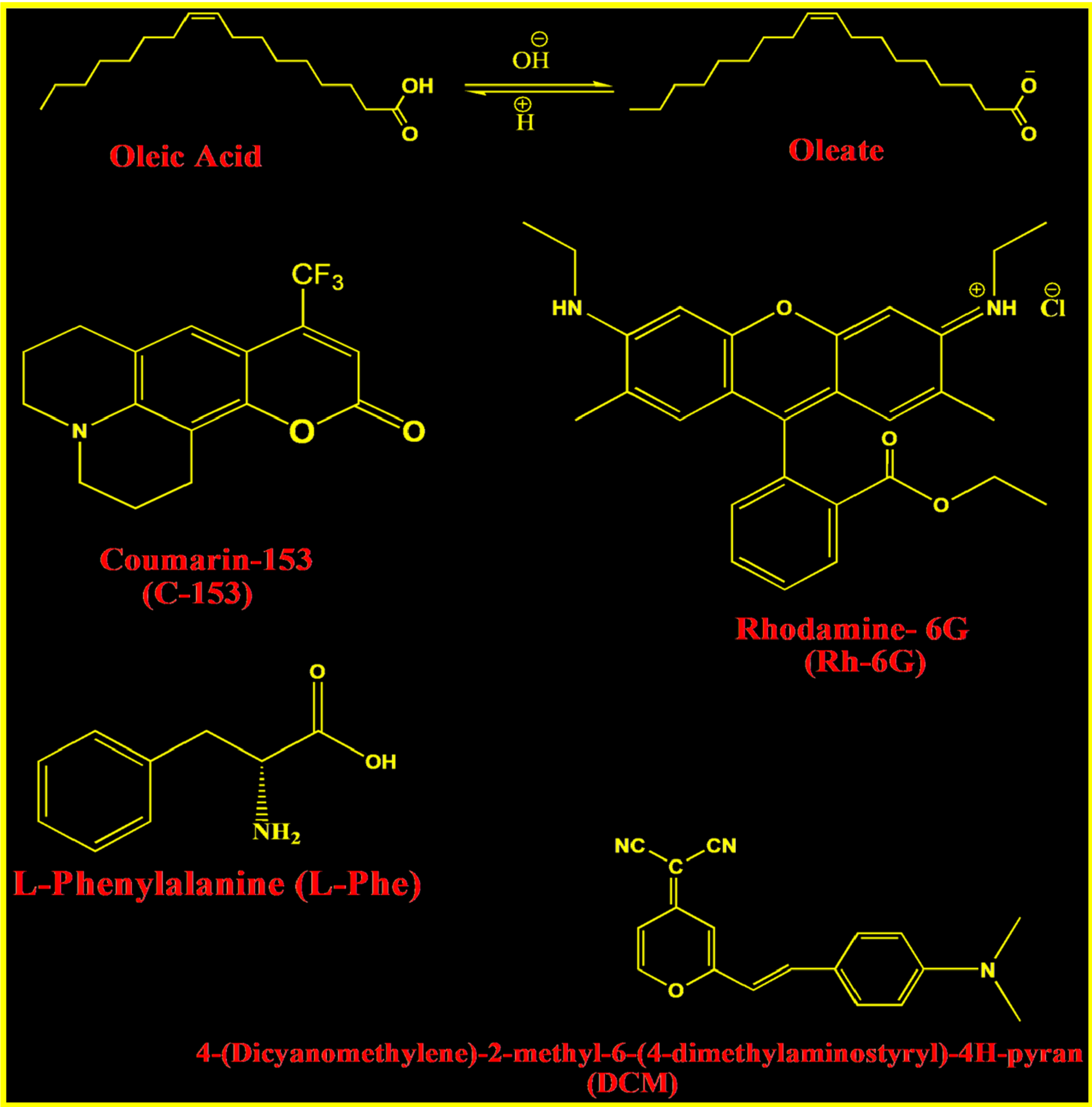
bilayer vesicles followed by an investigation of the effect of the fibrils on such lipid membranes.[9−12] On the one hand, using such a mimicked bilayer model membrane composed of the 1,2-dipalmitoyl-*sn*-glycerol-3-phosphatidylcholine (DPPC) phospholipid, a major component of human cell membranes,[13] Perkins et al. investigated the effect of the Phe aggregates at the millimolar level of concentration and reported that Phe makes the model membrane much more permeable, which has further been assigned as the underlying reason for several detrimental neurological disorders.[12] On the other hand, we recently found that this enhanced permeability of the model phospholipid membrane is also related to a decreased rigidity of the model bilayer phospholipid membrane as a result of the interaction of the Phe assemblies.[9] However, Phe not only affects the rigidity of the membrane but also has been reported to introduce damage to the thylakoid membrane.[14] A significant effort was made in this aspect to understand the molecular mechanism of the membrane interaction by the Phe assemblies. On the one hand, a hydrophobic interaction as a driving force was reported for such an interaction, where the phenyl rings of the Phe were reported to insert at a normal geometry with respect to the lipid membrane surface.[10] On the other hand, on the basis of the experimental conclusion based on the Langmuir trough protocol along with the simulation study, the probable location of such Phe aggregates have been reported by Griffith et al., where the most probable location of the aggregates has been reported to be at the buried region of the lipid monolayer, which in turn introduces microstructural alterations of the hydrophobic region of the membrane.[11] In the same line, De et al. recently reported that self-assembled amyloid aggregates of L-Phe can introduce a fusion of the phospholipid vesicle membrane.[15] Again, the self-assembly of the amyloidogenic morphology of amyloid-β (Aβ) has been very recently reported to be strongly bound to the lipid membrane, thereby altering the structural integrity of the membrane, a cause reported being associated with the cascade events leading to Alzheimer's disease.[16] In another advanced work, Maity et al. reported that such self-assembling amino acids, like L-Phe, can be functionalized with gold nanoparticles, which, with an effective interaction with the phospholipid vesicles, can form the lipid corona.[17] Taken together, the effort has the view to investigate the effect of Phe assemblies on the phospholipid membrane. However, an often-neglected but significant research gap is there regarding the effect of the Phe assemblies on the fatty acid-based vesicle membrane. The topic is supposed to be primitive considering the fact that protocellular membranes (early cell membranes) are hypothesized to form by simple fatty acids rather than phospholipids, and the justification of the hypothesis got a validation from the bilayer-forming ability of the simple fatty acids.[18] The importance of the fatty acid-based biomimetic compartmentalization has a significance in the sense that these vesicular self-assemblies are the best template to mimic the protocellular membrane, thereby founding its application in the "origin of life" research[19,20] because of the self-replication ability of the said vesicles.[21,22]

In this context, one of the most important supramolecular confinements of fatty acid, that is, fatty acid vesicles, are known to serve as a promising biomimetic model membrane[18,19,23] and are used as vehicles to transport drugs, proteins, or other biomacromolecules.[20,24] For example, octanoic acid, decanoic acid, oleic acid, etc. are known to form vesicular structures under proper experimental conditions.[21,25,26] These structurally trivial amphiphiles attract attention over phospholipids because of their unique features, such as an autocatalytic function,[27,28] dynamic response,[18] and a growing ability in the presence of more fatty acids,[21] making them an extremely interesting system for the investigation of their various photophysical behavior. Moreover, the single-chain amphiphile-composed simple fatty acid vesicles have a priority over the double-chain phospholipids because of the presence of a high concentration of nonassociated monomers in equilibrium with fatty acid vesicles.[18] The other significant character of these amphiphiles is the spontaneous formation of vesicles in aqueous media at a pH near the pK_a of the corresponding fatty acid.[21] The higher permeability of so-formed fatty acid membranes over a lipid membrane allows small molecules, like a nucleotide, to perform a transmembrane diffusion.[26] Among the various fatty acids studied, one of the most common fatty acids in nature, oleic acid (OA), is known to form several supramolecular confined structures depending on the concentration as well as temperature[27] and pH of the solution.[28] OA has a pK_a of ∼8;[29] hence, at a pH greater than 8 the major portion of the species is expected to remain in the ionized oleate state, whereas the protonated form will dominate at a pH below 8. Generally, OA exists as a micelle at pH > 10, as a vesicle at a pH between 7.5 and 9, and both as micelle and vesicle at pH ≈ 9−10. Earlier, Suga et al. reported the cone-shaped morphology of the sodium oleate molecules, where their hydrophilic headgroups have been mentioned to occupy a larger volume compared to their monoalkyl tails, thereby resulting in micellar structures at higher pH values.[28] Thus, the formation of various supramolecular confined architectures by the same compound at different pH of the media is of enormous importance and motivates us to explore the dynamics at their internal microenvironments as well as surfaces employing sophisticated spectroscopic experiments, like fluorescence correlation spectroscopy (FCS), a solvation dynamics study, a time-resolved anisotropy measurement of a probe molecule, and steady-state fluorescence measurements, along with the various microscopic characterizations. A solvation dynamics experiment reveals the change in dynamics of water molecules at the vesicle and micellar interfaces, whereas an FCS study was performed to explore the lateral diffusion of the probe molecule to understand the microenvironment of the assemblies. Further, the time-resolved anisotropy experiment helps to elucidate the internal structural rigidity of OA/oleate assemblies at various pH. On the one hand, taken together, the work merits an advance in terms of the comparative exploration of various dynamic properties of the OA/oleate assemblies formed at different pH. On the other hand, the self-assembled fibrillar morphologies of the essential amino acid L-phenylalanine (L-Phe), responsible for the inborn error of a metabolic disorder like PKU, is found to strongly modulate the dynamics properties of the oleic acid vesicle membrane. Specifically, the interaction of the L-Phe assemblies with the oleic acid vesicle membrane reduces the rigidity of the vesicle membrane and alters the hydration properties of the membrane, as established from various spectroscopic and microscopic investigations.

2. METHODS

2.1. Materials. Oleic acid was purchased from Loba Chemie Ltd. and was used without any further purification. The dyes 4-(dicyanomethylene)-2-methyl-6-(4-dimethylaminostyryl)-4*H*-pyran (DCM), coumarin-153 (C-153), and rhodamine-6G (R-6G) were obtained from Exciton. These

Scheme 1. Composition of Various Chemicals Used in This Experiment[a]



[a]Oleic acid, 4-(dicyanomethylene)-2-methyl-6-(4-dimethylaminostyryl)-4*H*-pyran (DCM), and coumarin-153 (C-153), and rhodamine-6G (R-6G), L-phenylalanine (L-Phe).

dyes were used to perform steady-state fluorescence experiments, an FCS study, and solvation dynamics as well as anisotropy measurements and fluorescence lifetime imaging microscopy (FLIM) measurements. Sodium hydroxide (NaOH), hydrochloric acid (HCl), and L-Phe were bought from Sisco Research Laboratories Pvt. Ltd. (SRL). The chemical structures of the materials were shown in Scheme 1. Other associated chemicals used to perform various experiments were of analytical reagent grade. As per the requirement, doubly distilled Milli-Q water having a resistivity of 18.2 MΩ·cm at 298 K was used for the preparation of solutions. The temperature at which the experiments were executed was 298 K.

2.2. Preparation of Solutions. A standard literature protocol was followed to prepare the OA/oleate solution and the oleate solution at various pH values.[18] First 0.5 M aqueous NaOH and 20 mM OA solution were prepared separately, which were mixed in a proper volume to prepare an oleate solution at pH ≈ 11.77. The other two OA/oleate assemblies at pH ≈ 9.72 and ∼8.28 were prepared through the addition of the required volume of 1 M stock HCl to the previously prepared solution having pH ≈ 11.77. The micellar solutions at pH ≈ 9.72 and ∼11.77 were transparent and suitable for performing spectroscopic measurements; however, the vesicle solution of pH ≈ 8.28 was turbid (Figure 1a−d). Therefore, to perform the spectroscopic measurements, we sonicated the solution of vesicles of pH ≈ 8.28 by an ultrasonic probe sonicator having the frequency of ∼20 ± 3 kHz for 15 min (Processor SONOPROS PR-250 MP, Oscar Ultrasonics Pvt. Ltd.) followed by a centrifugation to remove the titanium particles introduced during the sonication, which produced a transparent solution (Scheme S1). For the preparation of the solution of L-Phe of a required concentration, the dry powder was weighed and added to Milli Q water to make the desired concentrations of the L-Phe solution, as mentioned before.[4,7] To capture the microscopic image of the fibrillar morphology of the L-Phe, a small volume of the L-Phe solution was drop cast and dried properly before the microscopic observation.

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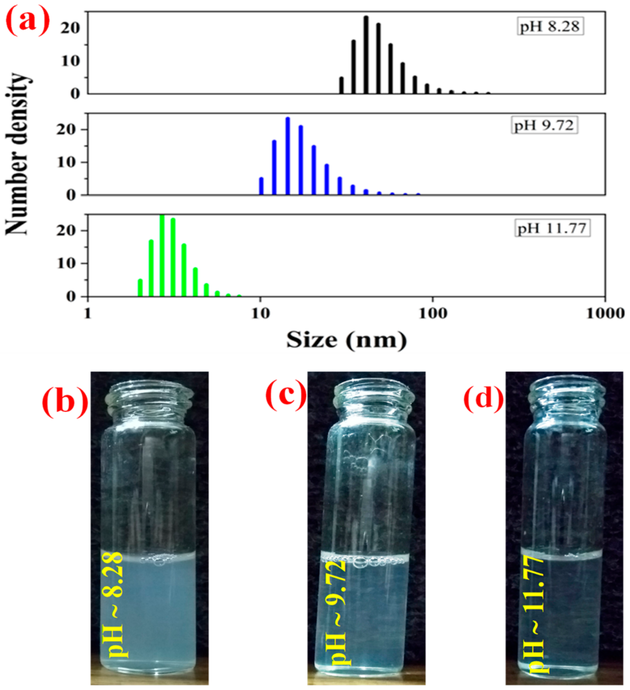

Figure 1. Number density-weighted size distribution profile of OA/oleate assemblies at different pH (a), as obtained from the DLS experiment. Images for various solutions of OA/oleate assemblies at pH ≈ 8.28 (b), pH ≈ 9.72 (c), and pH ≈ 11.77 (d) of the solution.

2.3. Steady-State and Time-Resolved Fluorescence Studies. On the one hand, the steady-state fluorescence emission spectra of DCM and C-153 within various fatty acid assemblies were captured using a spectrofluorimeter of Hitachi (model F-7000). On the other hand, for the time-resolved fluorescence emission of C-153 within the fatty acid assemblies, a picosecond time-framed time-correlated single-photon counting (TCSPC) setup was used. A description of this TCSPC setup having the instrument response function (IRF) of ∼100 ps was reported earlier.[30] Concisely, the excitation source was a picosecond diode laser (IBH, Nanoled) of 408 nm, and the emission decays were detected by a Hamamatsu microchannel plate photomultiplier tube (MCP PMT) (3809U) at the magic angle (54.7°) polarization. This was followed by an analysis of the time-resolved emission decays so obtained with the help of IBH DAS-6 decay analysis software.

Additionally, we measured the anisotropy decays of C-153 in the various fatty acid assemblies using the same TCSPC setup. During measurements, as an excitation source, a vertically polarized light was used, whereas to monitor the emission decays at parallel ($I_{\parallel}(t)$) and perpendicular ($I_{\perp}(t)$) polarizations, a motorized polarizer was placed. The experiment was continued until a specific peak difference between $I_{\parallel}(t)$ and $I_{\perp}(t)$ decays were achieved. In this context, the anisotropy decay function $r(t)$ was recalled as[31]

$$r(t) = \frac{I_{\parallel}(t) - GI_{\perp}(t)}{I_{\parallel}(t) + 2GI_{\perp}(t)} \tag{1}$$

In the above equation, G is a correction factor, the value of which is 0.6 for our instrument.

2.4. Fluorescence Lifetime Imaging Microscopy. FLIM is a well-known microscopic technique that provides an opportunity to record the image-based information for various macromolecular aggregates. The instrumental setup in our system consists of the laser scanning microscope of Becker & Hickl (DCS-120), which consists of the inverted type of optical microscope of Zeiss managed by a galvodrive component. Using this setup, first, the lifetime images of the samples were captured using two avalanche photodiode detectors (ID-Quantique ID100) by exciting the sample with a 488 nm laser of 10 mW with a repetition rate of 50 MHz. Here, the fluorescence light was separated from that of the excitation source through the use of long-pass filters (498 nm). In terms of a further specification of the image-capturing protocol, the images were captured by employing the polarized fluorescence transient using a TCSPC detection unit (Becker & Hickl SPC-152, PHD-400-N reference diode).[32] However, the associated intensity images were taken from the photons in all time channels of the pixels.

2.5. Fluorescence Correlation Spectroscopy. For the FCS study, the same instrumentation configuration as that of the FLIM was used except for certain operating parameters. Briefly, the sample was excited with the same picosecond diode laser of 488 nm operated in the continuous wave mode. During the experimentation, the sample was kept on a glass coverslip, and a 40× objective was used. In this experiment, the femtoliter ordered small effective optical volume was created, in and around which the used probe molecules were supposed to diffuse.[33] The diffusion results in the fluctuation of the fluorescence intensity, which can be time-correlated for expressing the normalized autocorrelation function $G(\tau)$[34,35] followed by fitting the traces so obtained using a three-dimensional (3D) diffusion model, as shown below.

$$G(\tau) = \frac{1}{N}\sum_{i=0}^{2} A_i\left(\frac{1}{1+\left(\frac{\tau}{\tau_D^i}\right)}\right)\left(\frac{1}{1+\frac{1}{\omega^2}\left(\frac{\tau}{\tau_D^i}\right)}\right)^{1/2} \tag{2}$$

In the above equation, N indicates the number of dye molecules within the observed volume, A_i is the fractional weighting factor for the ith contribution to the autocorrelation curve, τ_D^i denotes the diffusion time of the dye molecules, and τ indicates delay time or lag time. Not to mention, in this equation, the structure parameter (ω) is the ratio of longitudinal radii (l) to transverse radii (r), and the relationship between r and D_t is usually mentioned as

$$D_t = \frac{r^2}{4\tau_D} \tag{3}$$

In the above equation, the time through which the probe molecules diffuse within the observed volume (diffusion time) is represented as τ_D. Descriptions of other associated instruments employed in the work are given in the Supporting Information.

3. RESULTS AND DISCUSSION

3.1. Characterization of OA/Oleate Assemblies. *3.1.1. Dynamic Light Scattering (DLS) Experiment.* The size distribution histogram of various assemblies of oleic acid at the three pH values of the solution was recorded using a DLS measurement. The average diameter of the species obtained from this experiment was ∼108 and ∼19 nm at pH ≈ 8.28 and ∼11.77, respectively (Figure 1a). On the one hand, at pH ≈ 11.77 the number distribution plot does not contain any peak

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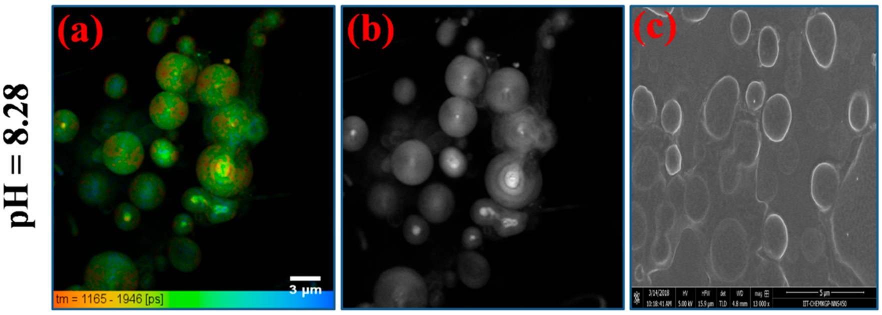

Figure 2. (a) DCM-stained fluorescence lifetime image of oleic acid assemblies at pH ≈ 8.28. (b) The intensity image of the corresponding lifetime images at pH ≈ 8.28. (c) FESEM image of oleic acid assemblies at pH ≈ 8.28.

at the higher distribution region, which essentially indicates the presence of the smaller particles, that is, the micelles. On the other hand, the average size at pH ≈ 8.28 was found to be 108 nm, the morphology of which was confirmed to be the vesicles, as observed from both of the field-emission scanning electron microscopy (FESEM) (Figure 2a,b) and FLIM images (Figure 2c). However, the DLS distribution histogram for the assemblies corresponding to pH ≈ 9.72 is worth correlating with the structural determination of these assemblies, where the average diameter was found to be ∼90 nm (Figure 1a). To elucidate the structures, we captured the transmission electron microscopy (TEM) images of the assemblies at pH ≈ 9.72 and ∼11.77. The presence of the spherical micelles at pH ≈ 11.77 is confirmed from the TEM image, as shown in Figure 3a,

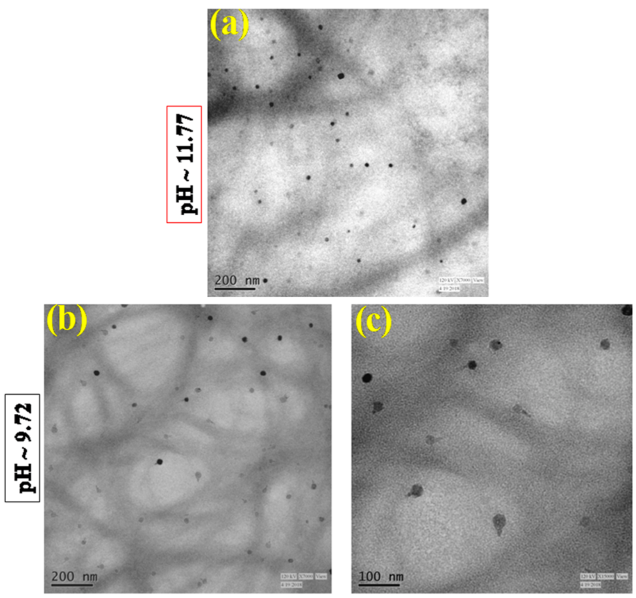

Figure 3. TEM images of the OA/oleate assemblies at pH ≈ 11.77 (a) and pH ≈ 9.72 (b). The enlarged image of (b) is shown in (c).

whereas the nonspherical nature of the micelles can be predicted at pH ≈ 9.72, as shown in Figure 3b and its enlarged image in Figure 3c. Not to mention, the structures at pH ≈ 9.72 were previously reported to be a cylindrical micelle, which is elongated in shape compared to the spherical micelle.[28] Therefore, the average diameter of 90 nm is supposed to correspond to the elongated micellar structures at pH ≈ 9.72.

3.1.2. Steady-State Fluorescence Emission Studies. We studied the steady-state emission spectra of the hydrophobic dye DCM in the OA/oleate assemblies at pH ≈ 8.28, 9.72, and 11.77. In the study, DCM was found to exhibit a strong emission band at ∼630 nm at pH ≈ 11.77 of the media, which was blue-shifted gradually with a decreasing of the pH of the solution, and ultimately became ∼625 nm at pH ≈ 8.28 (Figure 4a). This blue shift of the DCM dye in the solution having a low pH indicates a more hydrophobic environment of OA/oleate assemblies at pH ≈ 8.28 than at ∼11.77(Figure 4a). Also to a less extent, a similar type of blue shift of the fluorescence emission maxima was observed when the used probe was C-153 inside the OA/oleate assemblies at the above pH values (Figure 4b). Having the major location of these probes in the hydrophobic region of the assemblies, the blue shift of both the DCM and the C-153 mutually agrees with the increased hydrophobicity of the assemblies on moving from a high pH to a low pH of the solution.

Moreover, a change in the phase behavior of the solution was visually inspected when the pH of the solution was increased (Figure 1b−d). It was found that the solution was most turbid at pH ≈ 8.28 and least turbid at pH ≈ 11.77 (Figure 1b−d). The prominent turbidity of the solution at pH ≈ 8.28 indicates the formation of larger aggregates compared to the aggregates formed at the other two pH values of the solution studied. However, the justification of the alteration of the turbidity of the solution with the decrement of the pH of the solution can be made based on the size measurement of the assemblies as well as TEM observations of the so-formed assemblies at the various pH of the solution, as mentioned in the previous section.

3.2. Dynamic Properties of Various OA/Oleate Assemblies at Different pH of the Media. *3.2.1. Time-Resolved Fluorescence Anisotropy measurements.* We performed time-resolved anisotropy measurements using C-153 as a probe to interrogate the local organization of the OA/oleate assemblies at various pH values. This study is superior in understanding the extent of confinement of the local environment surrounding the probe molecule[36,37] in the assemblies. Here, the used probe C-153 to study the time-resolved fluorescence anisotropy measurements in the assemblies is mainly located near the hydrophobic region at the interface of the assemblies,[38] and hence the decay dynamics of the anisotropy of the probes provides information regarding the extent of the confinement of the microenvironment, where the probe molecules remain. The obtained anisotropy decay dynamics of C-153 inside various OA/oleate assemblies at different pH values were presented in Figure 5b, and the corresponding decay parameters were tabulated under Table 1. The anisotropy decay transients were found to be

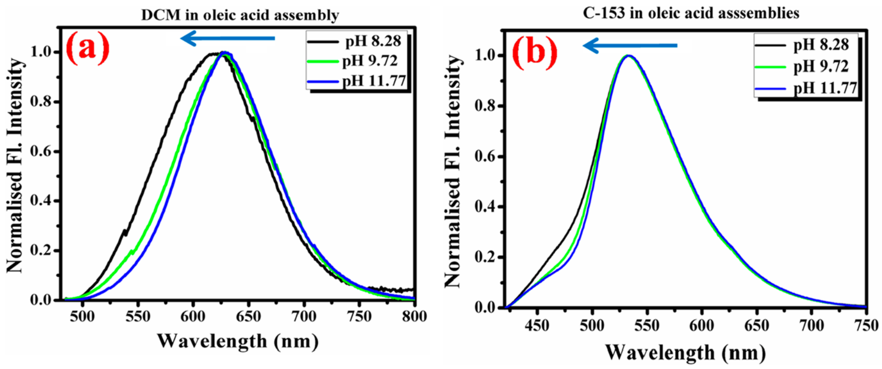

Figure 4. Steady-state fluorescence emission spectra of DCM (λ_{ex} = 480 nm) (a) and C-153 (λ_{ex} = 400 nm) (b) in various OA assemblies at different pH. The blue-colored arrows in both panels indicate the blue-shifting of the emission maxima of the probes when the pH is moved from 11.77 to 8.28.

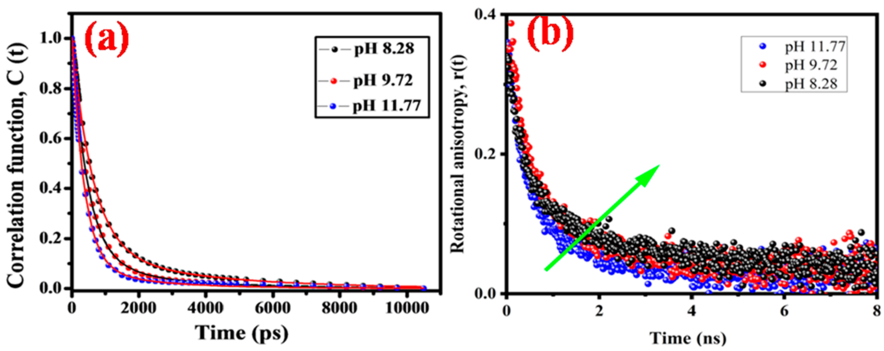

Figure 5. (a) Picosecond time-scaled solvent correlation function, $C(t)$ decay plot of C-153 in OA assemblies at various pH values (λ_{ex} = 400 nm). The points denote the actual value of $C(t)$, and the solid line denotes the best fit to a biexponential decay. (b) Picosecond time-scaled fluorescence anisotropy decays of C-153 in OA assemblies at various pH values (λ_{ex} = 400 nm).

Table 1. Parameters Obtained from Time-Resolved Anisotropy Decay Studies of C-153 in Oleic Acid Assemblies at Different pH (λ_{ex} = 400 nm)

pH	r_0	τ_{1r} (ps)	a_1 (%)	τ_{2r} (ps)	a_2 (%)	τ_{avg}[a] (ps)	χ^2
8.28	0.35	285	47	1872	53	1126	1.13
9.72	0.39	262	52	1716	48	960	1.15
11.77	0.37	274	53	1529	47	863	1.14

[a]Experimental error ±5%.

well-fitted with the biexponential fitting function, which indicates the orientational preference of the C-153 molecules in terms of the partition between the interfacial region and the hydrophobic region of both the micelle and vesicle. Not to mention, on the one hand, the fast component on the order of 0.2 ns is due to the free rotation of the probes in the interfacial water, which is almost the same in all the OA/oleate assemblies. On the other hand, from the decay parameters obtained after the fitting (Table 1), it is seen that, in the vesicle (pH ≈ 8.28), the biexponential anisotropy decay is composed of a longer time constant of ~1872 ps (53%) and a shorter time constant of ~285 ps (47%), thereby giving rise to an average rotational time of C-153 ≈ 1126 ps. This average rotational time in an elongated micelle (pH ≈ 9.72) and spherical micelle (pH ≈ 11.77) drops down to ~960 and ~863 ps, respectively. Interestingly, the time length and the contribution of the slow component becomes longer from 1529 ps at pH ≈ 11.77 to 1872 ps at pH ≈ 8.28, which is because of a more hindered rotation of the probe inside the more hydrophobic region of the vesicle (pH ≈ 8.28) than micelle (pH ≈ 11.77). This is because, at pH ≈ 8.28, the vesicular bilayer aggregates are present, whose nonpolar hydrocarbon chains are well-organized and tightly packed, thereby producing a rigid environment compared to the elongated or normal spherical micelles at the other two reported pH values. Hence, the more restricted rotation of C-153 in the vesicles compared to the micelle is due to the rigid spatial confinement imposed by vesicles (pH ≈ 8.28) compared to micelles (pH ≈ 9.72 and pH ≈ 11.77). Ghosh et al.[39,40] also reported a similar trend in anisotropy change between the micelle and vesicle formed of various surfactants and surfactant-like ionic liquids, where the more rigid packing of bilayer content in the vesicle is said to be responsible for the restricted rotation of the probe molecule compared to that in the micelle. The same probe C-153 was used by Mondal et al.[38] to interrogate the anisotropy change between micelle and vesicle, and a similar conclusion was reported. Additionally, since the slow component (τ_{2r}) notably decreases with an increase of the pH, there is a possibility that the motion of the probe molecule is associated with the global motion of assemblies at different pH. Thus, we observe that, with the decrease in size, the τ_{2r} becomes faster. Overall, the time-resolved anisotropy measurements indicate that, for the OA/oleate assemblies studied at the three pH values, the assembly at pH ≈ 8.28 has the characteristics of the highest internal extent of confinement compared to the other two assemblies at the two pH values studied.

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3.2.2. Solvation Dynamics Study using C-153 as a Probe. We extended our experimentation to track the water dynamics in and around the various OA/oleate assemblies at different pH values by performing a picoseconds time-framed solvation dynamics experiment, which is extremely sensitive to track the dynamics of water in a large number of confined media, such as vesicles, micelles, and also in many biological systems.[41−46]

Now, in studying the solvation dynamics, it is customary to build time-resolved emission spectra (TRES) using best-fit parameters of the emission decays[39,47] along with the steady-state emission spectra of the probe, as described by Fleming and Maroncelli.[48] This solvation dynamics process is usually expressed by the solvent correlation function $C(t)$, which is designated as

$$c(t) = \frac{\nu(t) - \nu(\infty)}{\nu(0) - \nu(\infty)} \tag{4}$$

where $\nu(0)$ denotes the peak frequency at zero time, $\nu(t)$ at time t, and $\nu(\infty)$ at infinite time. The decays obtained for $C(t)$ can be fitted applying the following multiexponential function.

$$C(t) = \sum_{i}^{n} a_i \exp(-t/\tau_i) \tag{5}$$

The average solvation time ($\langle\tau_s\rangle$) can be estimated by employing the following equation.

$$\langle\tau_s\rangle = \sum_{i}^{n} a_i\tau_i \tag{6}$$

a_i in the above equations indicates the relative amplitude of solvation, and τ_i stands for corresponding solvation times.

The $C(t)$ versus time plot is shown in Figure 5a, and the corresponding decay parameters were tabulated in Table 2,

Table 2. Picosecond Time-Scaled Solvent Correlation Function, $C(t)$ Decay Parameters of C-153 in Oleic Acid Assemblies at Various pH Values (λ_{ex} = 400 nm)

pH	τ_1 (ps)	a_1	τ_2 (ps)	a_2	τ_{avg}[a] (ps)
8.28	617	0.84	3354	0.16	1054
9.72	461	0.87	2322	0.13	702
11.77	363	0.91	2000	0.09	510

[a]Experimental error ±5%.

where $C(t)$ was found to decay biexponentially, indicating the presence of two types of water molecules in and around the OA/oleate assemblies at various pH. The slow dynamics are because of the formation of strong hydrogen bonds between ionic head groups at the interfaces and the nearest water molecules ("bound water"). For the vesicle structures at pH ≈ 8.28, the solvation time due to this bound water is 3.3 ns. The other type of water molecules, that is, the "exchange water", continuously exchanges hydrogen bonds between bound and unbound water,[49] and the solvation time due to this exchange water is 0.6 ns for the vesicle. Now, for a micelle at pH ≈ 11.77 the fast and slow time components of solvation are 0.3 and 2 ns, respectively, whereas for an elongated micelle these components are 0.4 and 2.3 ns, respectively. Earlier, Kumbhakar also proposed biexponential solvation time constants on the order of 2 and 0.2 ns in micelles.[50] It is reported that the sluggish solvation time component in the micelle is mostly governed by the rate at which the mechanically trapped and thermodynamically bound water molecules near the probe are exchanged,[51] whereas the fast time component is majorly determined by the cumulative response of such water molecules that are positioned relatively away from the probe.[52] Overall, it turns out that the average solvation time of ∼1054 ps happens in the case of the vesicle (pH 8.28), ∼510 ps in the case of the micelle (pH ≈11.77), and an intermediate time scale of ∼702 ps when the OA/oleate assemblies are at pH ≈ 9.72. So, the solvation dynamics of C-153 are found to be the slowest in the vesicle at pH ≈ 8.28 among all three assemblies studied at various pH. This suggests that the water molecules in the vesicles at pH ≈ 8.28 are more constrained compared to the micelles at the other two pH values. Again, the time scales of solvation dynamics at an interim pH of ∼9.72 indicate the presence of more constrained water molecules than the micelle but less constrained than the vesicle.

3.2.3. Fluorescence Correlation Spectroscopy Study to Investigate the Lateral Diffusion of Probe Molecules Inside OA/Oleate Assemblies. To understand the translational diffusion of probe molecules inside the so-formed confined media of OA/oleate assemblies at various pH, an FCS experiment was performed. This experiment experienced a renaissance in the 1990s,[53] and still, it holds its importance to study single fluorescently labeled molecules inside various organized media.[54−56] To study the translational diffusion of the probe molecules inside the confined media, first a focal laser beam was used to create a tiny confocal volume, within which the probe molecules were excited followed by the burst of fluorescent photon production. Now, the observed

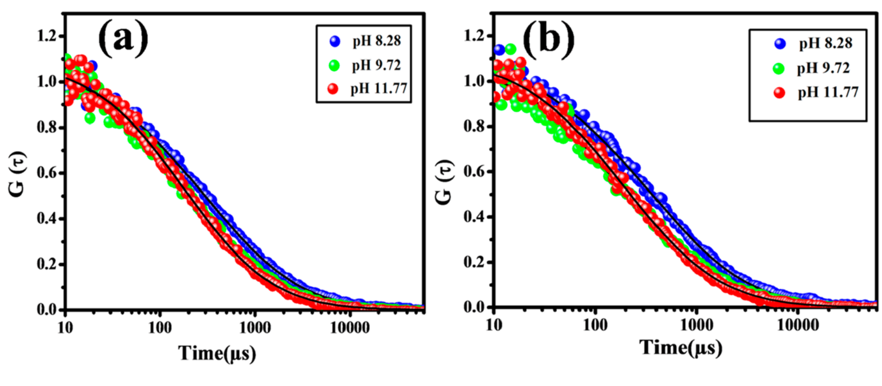

Figure 6. Fluorescence autocorrelation curves of (a) DCM and (b) R-6G in various oleic acid assemblies at different pH values. The points show the experimental decays of $G(\tau)$, and the solid line (black) indicates the best fit.

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fluorescence intensity was fluctuated because of the rapid Brownian motion of the probe molecules, which forces them to get in and out of the effective confocal volume. This fluctuation in the fluorescence intensity provides information about the diffusion of the probe molecules inside various confined media, which in turn helps to elucidate a certain extent of confinement of the assemblies within which the probe molecules diffuse. This extent of diffusion certainly depends on the surrounding environment around the probe molecules,[57] which is generally ascribed to the numeric value of diffusion coefficient D_t. Here we monitored the diffusion of two fluorescence probes, hydrophobic DCM and hydrophilic rhodamine-6G (R-6G) inside each OA/oleate at three pH values. The autocorrelation traces so obtained after the FCS experiment was performed were shown in Figure 6a,b, and the corresponding diffusion parameters are depicted in Tables 3 and 4. Being a hydrophobic dye, mostly the DCM molecules will present at the interfacial region of surfactant bilayer in the case of the vesicle (pH ≈ 8.28) and at the interfacial corona region in the case of micelles. The diffusion coefficient D_t of DCM in the vesicle solution is ~54 $\mu m^2\ s^{-1}$, which increases to ~90 $\mu m^2\ s^{-1}$ in the micelle at pH ≈ 11.77, thereby indicating a decreased trend of the diffusion time (τ_D) of DCM on moving from the vesicle at pH 8.28 to the micelle at pH 11.77 (Table S1, Supporting Information). The smaller value of D_t in the vesicle compared to the micelle indicates a more restricted rotation of DCM in the vesicle compared to the micelle. Because of the more symmetrical and tight packing of the hydrophobic tails in the vesicle, the internal organization is more confined in the vesicle than in the micelle. On the one hand, this rigid confinement restricts the diffusion of the DCM inside the bilayer region, which is the reason for the smaller value of D_t in the vesicle. On the other hand, the translational diffusion of the hydrophilic cationic probe R-6G was also tested within those assemblies. In neat water, the translational diffusion coefficient of R-6G was found to be 426 $\mu m^2\ s^{-1}$,[58] which was decreased in these assemblies because of the special confined environments formed by OA/oleate assemblies. On the one hand, in the case of diffusion of the molecules in the micelles at pH ≈ 11.77, the D_t value was found to be ~4 times less compared to the D_t value in net water (Table 4), which indicates the incorporation of the molecules into the micellar environment. On the other hand, the magnitude of D_t was found to be further reduced in the vesicle assembly at pH ≈ 8.28 compared to the D_t values in the micellar assemblies at pH ≈ 9.72 and ~11.77, thereby indicating a decreased trend of the diffusion time (τ_D) of R-6G on moving from a vesicle at pH ≈ 8.28 to a micelle at pH ≈ 11.77 (Table S2, Supporting Information). This is simply because of the extended confinement and rigid packing of the bilayer constituents in the vesicle. That is why the D_t value was found to be appreciably less in vesicles compared to the D_t values in the other two micellar assemblies. Therefore, this reduced magnitude of the D_t value in the vesicular assemblies at pH ≈ 8.28 justifies the hindered diffusion of the probes because of the large extent of confinement in these vesicular assemblies compared to that of the micellar assemblies at the other two pH values.

Table 3. Lateral Diffusion Coefficient, D_t Values of DCM in Various Oleic Acid Assemblies at Different pH Values

pH	$(D_t)_{\text{OA assemblies}}$ ($\mu m^2\ s^{-1}$)	$a_{\text{assemblies}}$[a] (%)
8.28	54.5	64
9.72	58.6	43
11.77	90.7	56

[a]Experimental error ±5%.

Table 4. Lateral Diffusion Coefficient, $D(t)$ values of Rhodamine-6G in Various Oleic Acid Assemblies at Different pH Values

pH	$(D_t)_{\text{OA assemblies}}$ ($\mu m^2\ s^{-1}$)	$a_{\text{assemblies}}$[a] (%)
8.28	52	56
9.72	63	39
11.77	100	53

[a]Experimental error ±5%.

3.3. Interaction of the L-Phenylalanine (L-Phe) Assemblies with the Fatty Acid Vesicle Membrane.

3.3.1. Microscopic Characterization of the L-Phe Assemblies. To investigate the effect of the assemblies of L-Phe on the vesicle membrane, first, we characterized the architectures formed by the L-Phe. Three net concentrations of L-Phe were chosen, namely, 1.2, 5, and 12 mM. The signatures of the fibril formation at all of these concentrations were validated by using fluorescence lifetime imaging microscopy with DCM as a dye, atomic force microscopy (AFM), and field emission scanning electron microscopy. The fibrillar assemblies at 1.2 mM concentration of L-Phe were shown in Figure 7a−c, which represents the FLIM, FESEM, and AFM investigations, respectively. The corresponding observations for the 5 mM L-Phe were presented in Figure 7d−f, whereas the morphologies at 12 mM L-Phe were presented in Figure 7g−i. Therefore, a primary observation of the morphologies formed by the L-Phe at the experimental concentrations revealed the formation of the fibrillar morphologies. However, the in-depth analysis of the FLIM images provides additional information regarding differences in the properties of the fibrils so formed at different concentrations. The advantage of this FLIM technique is associated with the sensitivity of the lifetime of the used probe toward the surrounding environment of the assemblies where they remain, where it provides the capturing of an image of the various supramolecular aggregates in one hand, and on the other hand, based on the analysis of the time scales associated with the captured images, various important properties of the assemblies, like, rigidity or fluidity of the assemblies, the heterogeneity of the assemblies, etc. can be predicted.[32,59] Here, the lifetime distribution histogram corresponding to the FLIM images of the fibrils at different concentrations was analyzed and presented in Figure 8. The same technique was earlier reported by Banik et al. to characterize various properties of the L-Phe fibrils, where the lifetime of the probe inside the fibrils was taken as the marker of the environmental alterations of the fibrils.[7,60] In our present study, it was found that the lifetime distribution histogram corresponding to the FLIM images shows an appreciable difference in changing the concentration of the L-Phe fibrils. For example, the distribution histogram of the DCM in the 1.2 mM fibrils shows an all area-average lifetime distribution ranges between 430 and 2624 ps (Figure 7a). On the one hand, the corresponding lifetime distribution to this Figure 7a is shown in Figure 8a, where the Gaussian fit of the distribution histogram provides the value of the central lifetime of 1488 ps. On the other hand, the distribution histogram of the DCM in the 5 mM fibrils shows an all area-average lifetime distribution

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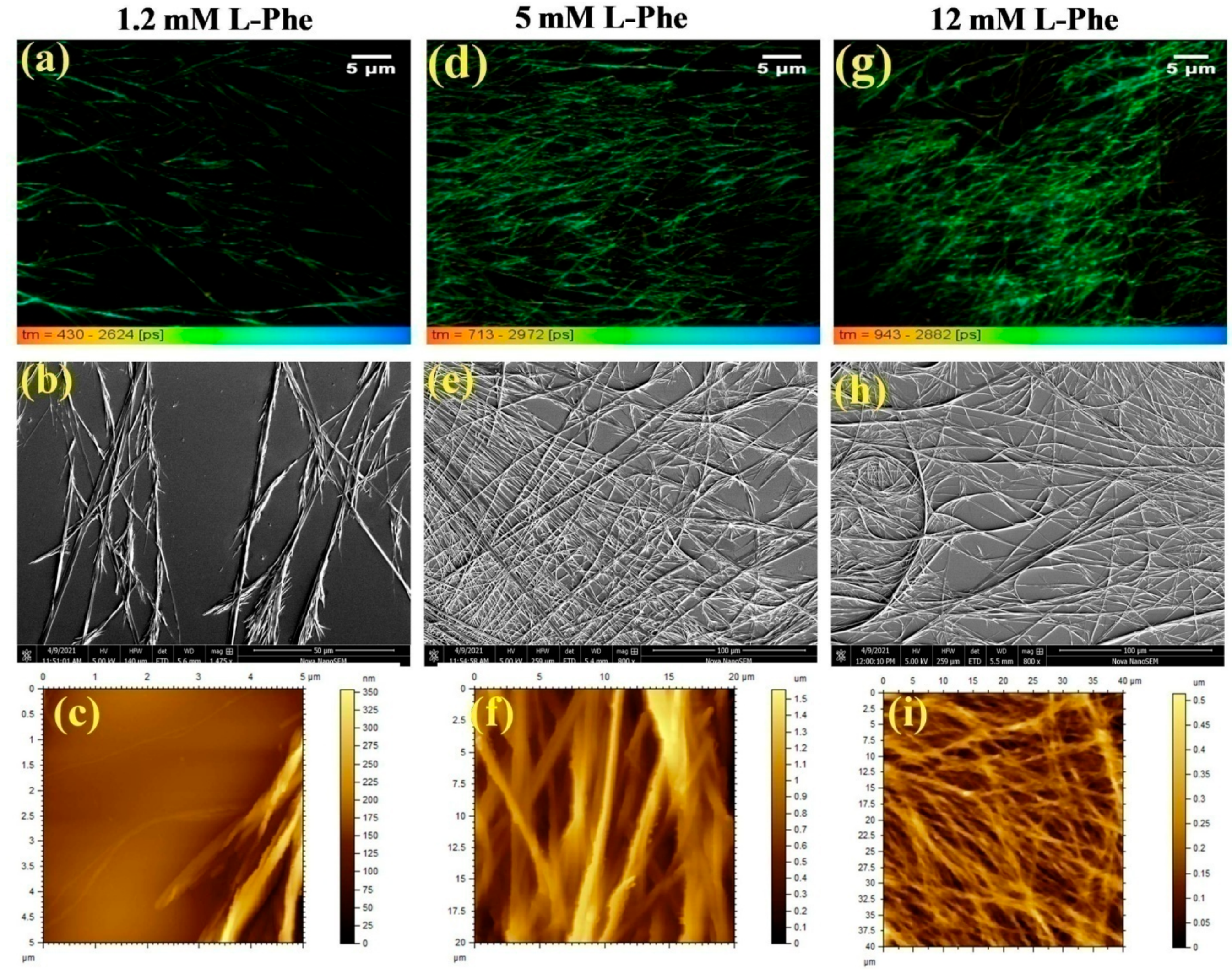

Figure 7. Microscopic observations for the fibril formation by various concentrations of L-Phe, as observed from FLIM (a, d, g), FESEM (b, e, h), and AFM (c, f, i) for 1.2, 5, and 12 mM L-Phe solutions, respectively.

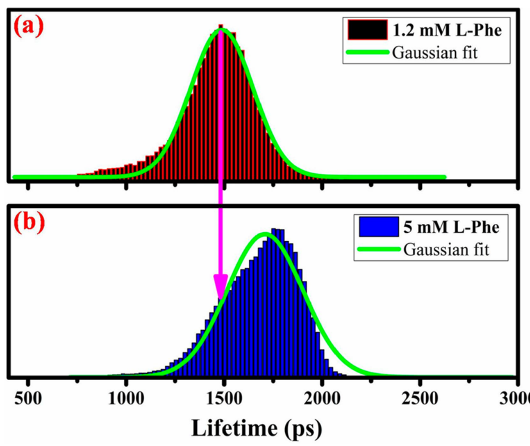

Figure 8. Lifetime distribution histogram of DCM dye inside fibrillar assemblies formed by (a) 1.2 mM L-Phe and (b) 5 mM L-Phe.The pink arrow indicates the shifting of the central lifetime value of DCM from 1.2 to 5 mM L-Phe after the histograms were fitted with the Gaussian function.

ranges between 713 and 2972 ps (Figure 7d). The corresponding lifetime distribution to this Figure 7d is shown in Figure 8b, where the Gaussian fit of the distribution histogram estimates the value of the central lifetime of 1708 ps. This signature of the elevated shifting of the lifetime maxima value (Figure 8) indicates that the global assemblies in the case of 5 mM L-Phe provide a much more rigid environment as a whole compared to the global environment provided by the fibrillar assemblies at 1.2 mM L-Phe. The probable reason for the enhanced rigidity of the assemblies at the higher concentration of L-Phe is supposed to be composed of more number monomer units, which, by their global assembly formation, form a more rigid environment compared to the assemblies formed at the lower concentration of the L-Phe.

3.3.2. Alteration of the Size of the Vesicle Membrane Due to Interaction with L-Phe Fibrils. To investigate the effect of the L-Phe assemblies on the oleic acid vesicle membrane, we measured the size distribution histogram of the vesicles using the DLS technique, and the corresponding histograms are provided in Figure 9a. In the case of the neat sonicated OA vesicles, the average diameter of the vesicles was ∼ 95 nm, which was found to increase slightly in the presence of the added L-Phe to the vesicles in a concentration-dependent manner (Figure 9a). In addition to this, a close observation of such histograms indicates that, along with the increment of the

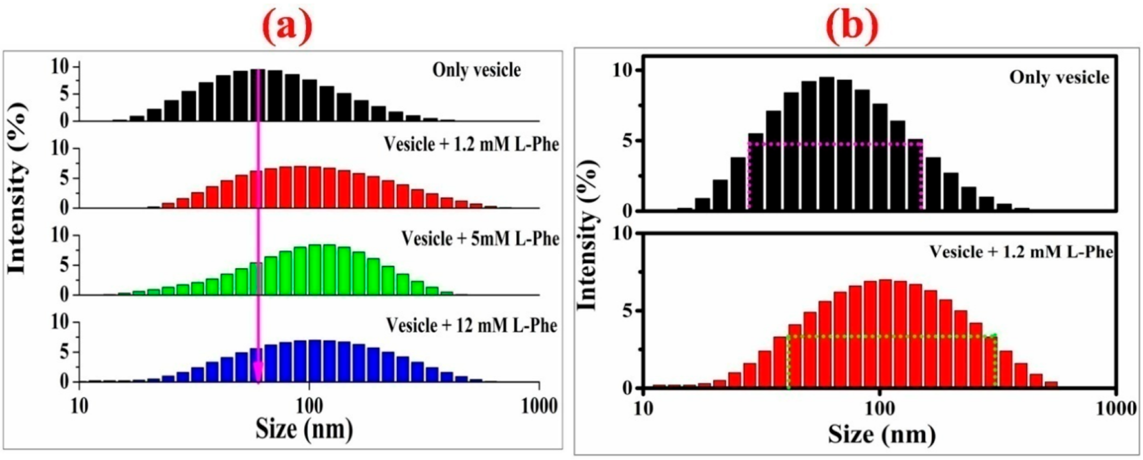

Figure 9. (a) The size distribution histogram of the OA vesicles in the absence and presence of various concentrations of L-Phe. The pink arrow guides to the shifting of the histogram intensity maxima to the right-hand side on increasing the concentrations of L-Phe to the OA vesicles. (b) The alterations of the fwhm of the histogram for the OA vesicles in the absence and presence of 1.2 mM L-Phe. The corresponding fwhm lines are presented with the pink and green colored lines on the histograms. The resultant histograms are obtained via the triplicate mode of measurements. [OA vesicles] = 10 mM, [L-Phe] = 1.2−12 mM. Error = ±3%.

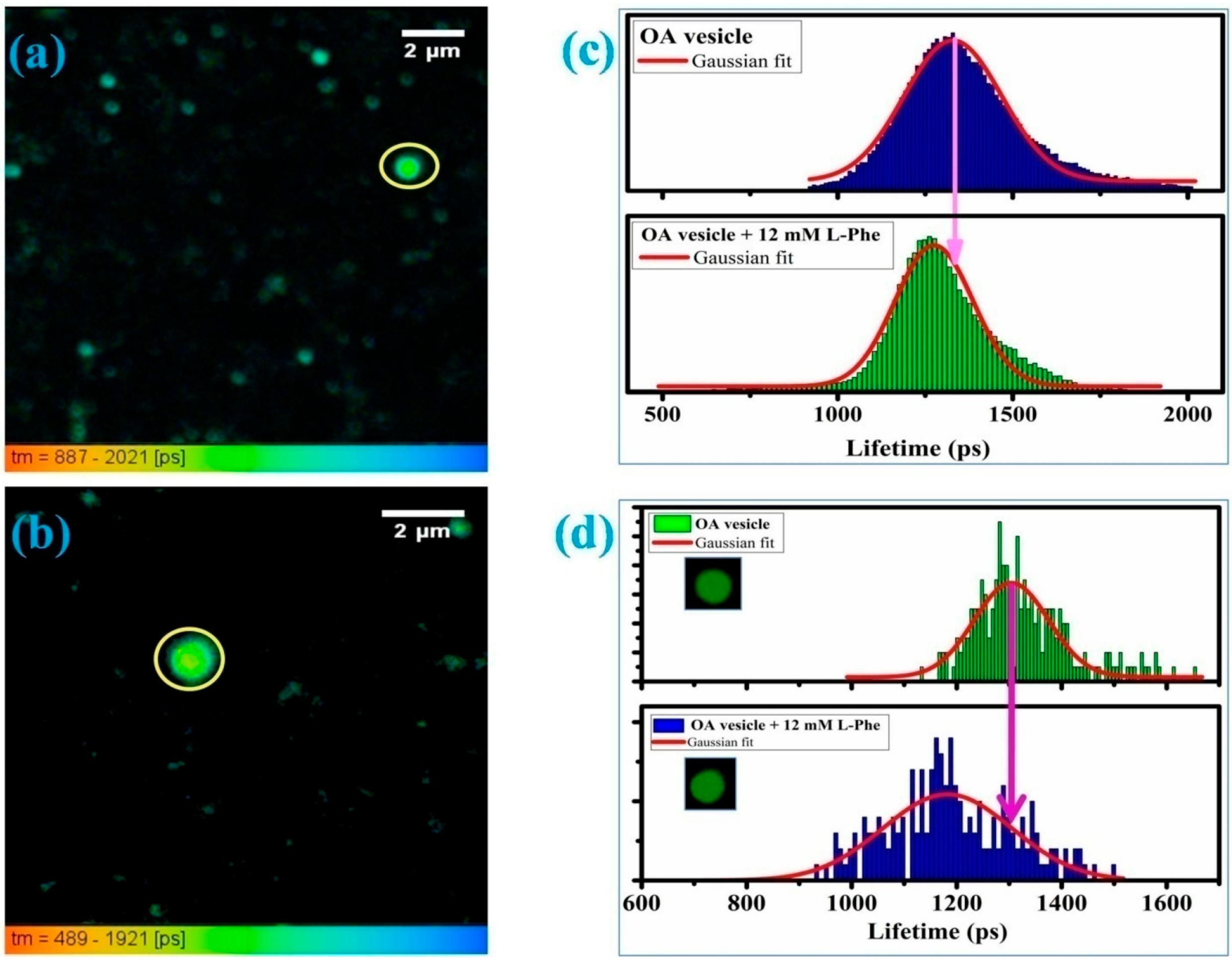

Figure 10. Fluorescence lifetime imaging microscopic investigation using DCM as a marker for the neat vesicles (a) and vesicles in the presence of 12 mM L-Phe. The corresponding global analysis for the histogram of the lifetime distribution is presented in (c). The lifetime distribution histogram corresponding to the single vesicles, as circled with a yellow line (a, b), is shown in (d), where the pink arrow indicates the shift of the central lifetime value after being fit to the Gaussian function, and the corresponding single vesicles are shown in the inset (d).

size of the vesicles, the size distribution becomes broader in the presence of the L-Phe fibrils. For example, in the case of the neat vesicles, the full-width half-maxima (fwhm) of the distribution was found to be 123 nm, which increased to

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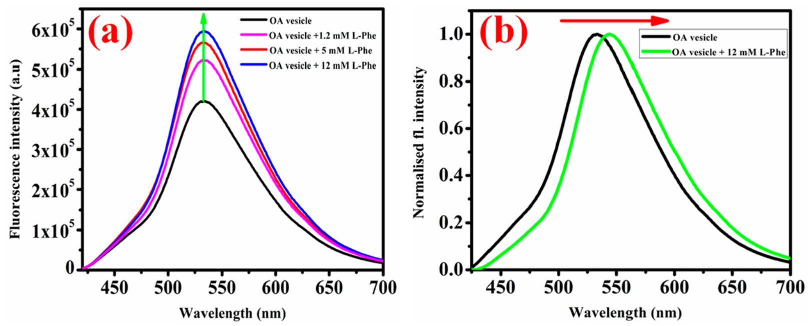

Figure 11. Unnormalized (a) and normalized (b) fluorescence emission spectra of C-153 in the OA vesicles in the absence and presence of L-Phe. (a) The green arrow indicates the enhancement of the fluorescence intensity in response to the added L-Phe to the vesicles. (b) The red arrow indicates the red-shift of the emission maxima of C-153 in the L-Phe-containing OA vesicles compared to neat OA vesicles. [OA vesicles] = 10 mM, [L-Phe] = 1.2−12 mM, [C-153] = 1×10^{-4} M. Error = ±3%.

267 nm in the presence of 1.2 mM L-Phe to the membrane (Figure 9b). On the one hand, this increased fwhm of the distribution histograms of the membranes in the presence of the L-Phe fibrils indicates that the vesicles become much more heterogeneous as a result of the interaction with the Phe fibrils. On the other hand, for the increment of the average diameter of the vesicles in the presence of the fibrils, an indication of the possibility of the insertion of the L-Phe to the membrane bilayer region emerges, which on one hand increases the heterogeneity of the membrane bilayer, as observed from the increased value of the fwhm of the size distribution histogram, and on the other hand, such an insertion of the Phe is expected to loosen the membrane bilayer packing, the result of which might be the loss of bilayer integrity, which in turn is supposed to make the bilayer region of the membrane much more flexible.

3.3.3. Characterizations of the Vesicle Bilayer Flexibility in the Presence of L-Phe using Fluorescence Lifetime Imaging Microscopy. To validate the hypothesis regarding this increased flexibility of the membrane in the presence of the added L-Phe, a fluorescence lifetime imaging microscopic investigation followed by an image analysis was performed. Not to mention, for the imaging, the probe DCM was incorporated into the vesicles, since it has been reported to be a marker of the membrane bilayer.[61] The FLIM images of the neat vesicles and that in the presence of the 12 mM L-Phe were presented in Figure 10a,b, respectively. A close observation of the lifetime images indicates no structural alterations of the vesicles as such; however, an in-depth analysis of the corresponding lifetimes to each of these images indicates that the membrane properties are altered significantly in the presence of L-Phe compared to the neat vesicles. The global lifetime distribution histogram corresponding to the neat vesicles under the entire magnification of the sample on Gaussian fitting (Figure 10c) estimates the central lifetime of the DCM dye as 1329 ps, which decreases to 1275 ps in the case of the addition of 12 mM of L-Phe to the vesicles. This shift of the lifetime value of DCM toward the lower value (1275 ps) in the L-Phe-containing vesicles (Figure 10c) indicates that the rigidity of the membrane decreases in the presence of L-Phe. Earlier, we reported the elevated membrane fluidity of the lipid membranes in the presence of externally added peptides or neurotransmitters based on the decreased value of the lifetime of the DCM based on the FLIM investigations.[62,63] Although the global analysis indicates an average decrement of the lifetime of all the vesicles under the given magnification, we further tend to probe a single vesicle from the entire magnification, as shown by yellow circles in Figure 10a,b, and the corresponding lifetime histogram to each of these single vesicles was presented in Figure 10d. From the histogram corresponding to the neat single vesicle (without L-Phe) (Figure 10d), the central lifetime of the DCM probe in this single vesicles was found to be 1304 ps after the Gaussian fitting, which decreases to 1182 ps when a single vesicle in the presence of the added L-Phe was considered. This quantifies the loss of the rigidity or the elevation of the fluidity of the single vesicle in the presence of 12 mM L-Phe, the estimation of which is much more rigorous compared to the average estimation, as made based on the histogram presented in Figure 10c. Taken together, the conclusion regarding the loss of the membrane bilayer rigidity is supposed to show a mutual coherence from either the lifetime analysis of the vesicles using the FLIM experiment or the time-resolved anisotropy experiment, as mentioned in the later part. In this aspect, the study by Perkins et al. regarding the effect of the Phe on a lipid membrane is worth mentioning, where they concluded that Phe introduces an elevated extent of permeability of the membrane.[8] This enhanced fluidity of the membrane in the presence of toxic assemblies of L-Phe might be associated with some of the chronic neurological disorders, like phenylketonuria.[4,60,64]

3.3.4. Spectroscopic Investigation to Probe the Alteration of the Properties of the Fatty Acid Membrane Due to Interaction with L-Phe Fibrils. 3.3.4.1. Steady-State Fluorescence Emission Study. In the given context, obvious questions are worth raising regarding the driving force responsible for such an interaction of the Phe assemblies with the model membrane. To answer the question, a steady-state fluorescence emission experiment was performed using the hydrophobic probe C-153 in the vesicles in the absence and presence of the L-Phe. Being a hydrophobic probe, the major location of C-153 is reported to be at the bilayer region of the membrane with a minor population toward the interface too.[65] This is supported by the fact that the fluorescence intensity and the quantum yield of the hydrophobic probes in the vesicle bilayer are much more compared to the fluorescence intensity of the probe in neat water.[66] For the maximum encapsulation of the probes in the vesicles, we took the probe to a vesicle concentration ratio of 1:100. Earlier, Bhattacharya et al. used a 1:80 ratio of the concentration of the hydrophobic probe to the vesicles in order to ensure the maximum encapsulation of the probes inside the vesicles.[66]

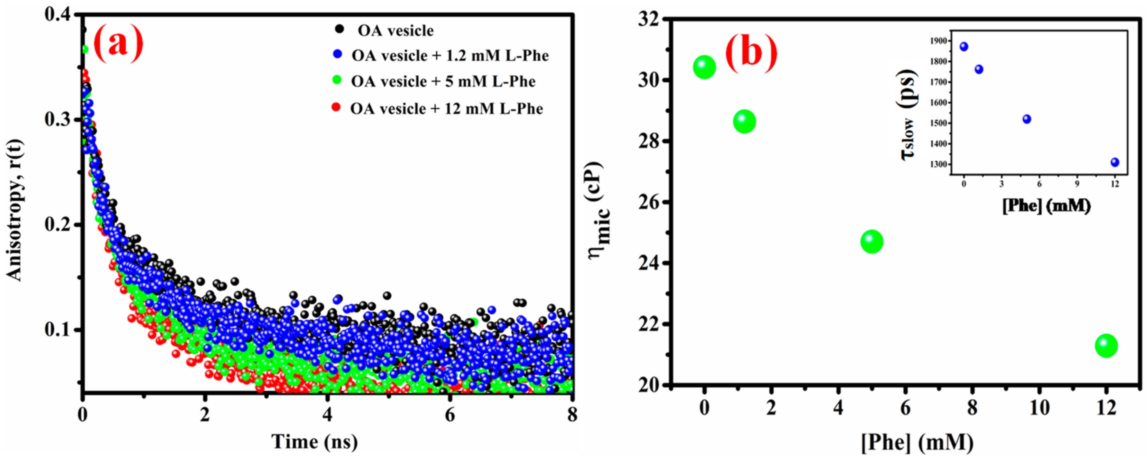

Figure 12. (a) Time-resolved anisotropy decay transients of C-153 in the OA vesicles in the absence and presence of different concentrations of added L-Phe. (b) Alterations of the microviscosity of the vesicle bilayer in the presence of added L-Phe. (inset) The relation between the slow component of rotation (τ_{slow}) of the probe with the added L-Phe to the vesicles. [OA vesicles] = 10 mM, [L-Phe] = 1.2−12 mM, [C-153] = 1 × 10^{-4} M. Error = ±3%.

However, the addition of the different concentrations of the L-Phe (1.2 to 12 mM) to the vesicles was found to elevate the emission intensity of the C-153 to ∼1.5 times in the presence of the highest concentration of the L-Phe added (Figure 11a) compared to the neat vesicles. This enhanced emission intensity of the hydrophobic probe inside the bilayer of the vesicles indicates that, in the presence of the L-Phe assemblies, the vesicle bilayer senses a more hydrophobic microenvironment in a concentration-dependent addition of L-Phe compared to the neat vesicles. Here, it is important to mention that, in the neat aqueous medium, Phe has been reported to form the architectures of olegomers[64] as well as fibrils.[4,64] The architectures so formed by the Phe were further characterized by Do et al. using ion-mobility mass spectrometry (IMMS) experiments, where the signature of a multiple layer formation of the Phe assemblies through the self-stacking of layers of monomers was reported.[64] There, the monomeric forms unite in a particular fashion such that the aromatic residues are exposed outside.[64] In terms of the interaction of such Phe assemblies with the model membrane, these hydrophobic aromatic residues of Phe facilitate the insertion of the species inside the model membrane.[10] Therefore, the reason for such an increment of hydrophobicity of the bilayer microenvironment is supposed to be associated with the insertion of the hydrophobic phenyl groups of the L-Phe assemblies, which aids the additional hydrophobicity to the hydrophobic bilayer region of the membrane. On the one hand, taken together, the hydrophobic interaction is one of the major driving forces for the insertion of the assemblies inside the bilayer region of the vesicles. On the other hand, in the presence of the high concentration of the L-Phe used (12 mM), the emission maxima of the probe was found to red-shift to 5 nm compared to the emission maxima of the probe in neat vesicles (Figure 11b). This is a signature of the increased permeation as well as an elevated destabilization of the vesicle bilayer in the presence of the L-Phe assemblies, which in turn can affect the rigidity of the membrane. Earlier, Banerjee et al. also utilized an ∼4 nm red-shift of the emission spectra of the hydrophobic probe inside the lipid vesicle bilayer in the presence of the peptide, which in turn was reported to be associated with the destabilization of the vesicle bilayer membrane.[62] On the one hand, overall, in the given context, the increased intensity of the emission spectra of C-153 inside the vesicle in the presence of L-Phe signifies the sensation of the elevated degree of the hydrophobic microenvironment of the bilayer. On the other hand, an ∼5 nm red-shift of the emission spectra indicates the possibility of the membrane destabilization of the vesicle bilayer in the presence of L-Phe. Not to mention, earlier work by Gerbelli et al. also did report the increased elasticity of the lecithin membrane in the presence of the fibril-forming unnatural amino acid diphenylalanine, which in turn elevates the flexibility of the membrane.[67] However, the steady-state fluorescence emission intensity (Figure S1a, Supporting Information) as well as emission maxima fluctuation of C-153 (Figure S1b, Supporting Information) in neat fibrillar solutions of L-Phe of various concentrations (1.2, 5, and 12 mM) was found to be insignificant compared to the emission behavior of the dye in the vesicles (Figure 11).

3.3.4.2. Time-Resolved Anisotropy Measurement. Now, to understand the alteration of rigidity of the membrane bilayer due to the interaction with the L-Phe fibrils at the different concentrations added, a reorientational dynamics of the C-153 probe was measured using the time-resolved anisotropy decay measurement, and the obtained anisotropy decay traces were presented in Figure 12a with the presentation of the corresponding fitting parameters in Table 5. The time constants obtained from this experiment provide the information regarding the rotational degree of freedom, and taking advantage of this, a number of literature reports are

Table 5. Time-Resolved Anisotropy Decay Parameters of C-153 in the Vesicles in the Absence and Presence of L-Phe

system	τ_1(ps)	a_1	τ_2(ps)	a_2	τ_{avg}[a] (ps)	χ^2
OA vesicle	1872	0.53	285	0.47	1126	1.03
OA vesicle +1.2 mM L-Phe	1762	0.55	270	0.45	1090	1.04
OA vesicle +5 mM L-Phe	1520	0.57	252	0.43	974	0.98
OA vesicle +12 mM L-Phe	1310	0.61	225	0.39	886	1.02

[a]Experimental error ±5%.

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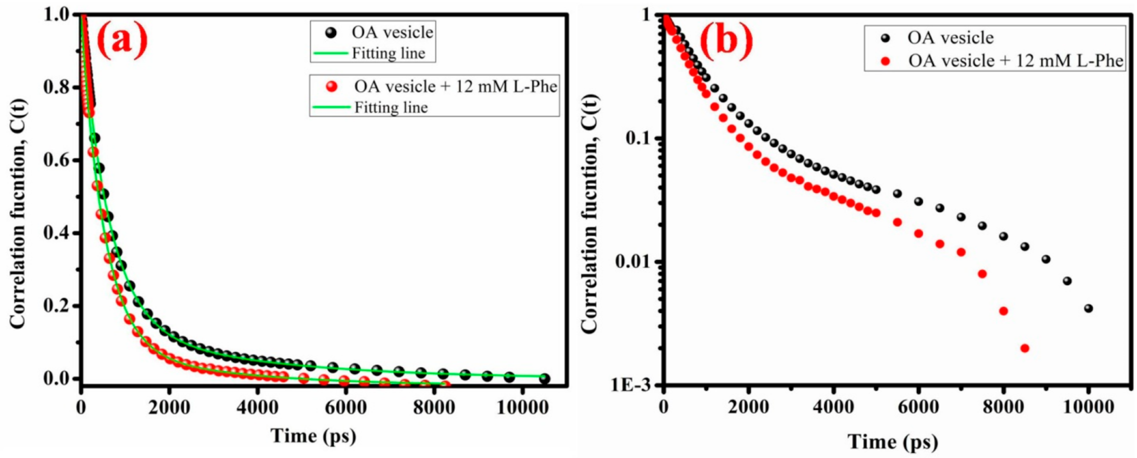

Figure 13. Decay of solvent correlation function $C(t)$ of C-153 in the OA vesicles in the absence and presence of L-Phe in exponential mode (a) and logarithmic mode (b). [OA vesicles] = 10 mM, [L-Phe] = 12 mM, [C-153] = 1×10^{-4} M. Error = ±3%.

there that reflect the extent of rigidity or the micro-heterogeneity of the vesicular assemblies.[36,65,68,69] The anisotropy decay fitting of the traces so obtained was found to consist of the two components. For example, in the case of the neat vesicles, the slow anisotropy constant of 1872 ps indicates the rotation of the probes inside the bilayer region of the vesicles, whereas the fast component of 285 ps indicates the rotation of the molecules at the interfacial region of the membrane. Because of the insertion of the assemblies at the bilayer region of the membrane, the corresponding slow component of rotation was found to be affected to the maximum extent. Interestingly, this slow component of rotation of the probes inside the bilayer region of the vesicles was found to be decreased in a concentration-dependent manner of the added L-Phe. The slow component of rotation becomes 1762 ps for the addition of 1.2 mM L-Phe, which further decreases to 1520 ps in the presence of 5 mM L-Phe and finally reaches 1310 ps in the presence of 12 mM L-Phe to the vesicles. However, the fast component of rotation was found to be decreased, but to a small extent, and the cumulative decreased contribution of both of the slow as well as the fast component of rotation leads the successive decrement of the average rotational time of the probe inside the vesicles with the successive increment of the added L-Phe. In the case of neat vesicles, the average rotational time was found to be 1126 ps, which decreases in a concentration-dependent manner of the added L-Phe and becomes 1090, 974, and 886 ps, in the presence of 1.2, 5, and 12 mM L-Phe to the vesicles, respectively. This decreased value of the slow component of the rotation and hence the average rotational time of the probe in the presence of the L-Phe indicates the greater extent of rotational freedom of the probe molecules inside the bilayer region of the vesicles in the presence of the added L-Phe, which in turn indicates the loss of rigidity of the membrane bilayer. In other words, the fluidity of the membrane is increased due to the interaction with the L-Phe fibrils. One probable reason for the loss of rigidity of the bilayer is the permeation of the Phe assemblies with the phenyl rings normal to the membrane bilayer, which is expected to loosen the structural integrity of the membrane bilayer, as a result of which the compact packing of the membrane decreases compared to the neat vesicle membrane without L-Phe. In the case of the model lipid membrane, Griffith et al. earlier reported the site of the location of the Phe inside the buried region of the membrane layer, which has further been reported to result in the microstructural alterations of the hydrocarbon region of the membrane.[11] Following the same conclusion, Nandi et al. also found the deformation of the membrane bilayer of model LAPC membrane due to the interaction with L-Phe, which decreases the rigidity of the membrane.[9] Hence, in the given context, the signature of the decreased rotational time constant of the probe inside the fatty acid vesicle membrane seems to follow an alike trend in terms of the decreased value of the rotational time for the probe inside the lipid model membrane in the presence of L-Phe. On the one hand, therefore, a greater rotational degree of freedom of the probe molecules inside the Phe-containing bilayer is the reflection of the insertion of the assemblies at the bilayer region of the membrane followed by the decreased extent of the confinement of the membrane bilayer. On the other hand, using the time-resolved anisotropy decay parameter corresponding to the rotation of the probe at the bilayer region of the membrane, we estimate one important parameter of the bilayer region of the membrane, which is the microviscosity of the bilayer of the membrane. Not to mention, the microviscosity is a more prominent parameter compared to the bulk viscosity for a heterogeneous system, like vesicles, which is composed of different environments, like, head region, interfacial region, bilayer region, etc. Because of the prominent impact of the assemblies of L-Phe on the membrane bilayer region, we include only the slow component of rotation of the probe to estimate the microviscosity of the membrane bilayer. The well-known Stokes−Einstein−Debye equation was used for the estimation of the microviscosity utilizing the value of the slow component of rotation of the probe, as obtained from the time-resolved anisotropy decay experiment. The said equation can be presented as mentioned below.

$$\tau_{\mathrm{slow}} = \frac{\eta_{\mathrm{mic}}}{KT} V \tag{7}$$

In expression 7, τ_{slow} indicates the slow component of rotation of the probe inside the bilayer region of the membrane, K is the Boltzmann constant, T is the experimental temperature (298 K), and V denotes the volume of C-153. In this context, the protocol of Edward's method[70] for the estimation of the volume of the probe estimates the volume of the C-153 as 253

Å^3.[68] Using this method, the microviscosity of the membrane bilayer of neat vesicles was found to be 30.42 cP. However, the value decreases steadily with the steady decrease of the slow component of the rotational motion of the probe inside the bilayer and finally reaches 28.64, 24.70, and 21.29 cP for the addition of 1.2, 5, and 12 mM L-Phe to the vesicles, respectively. A plot of the change of this microviscosity with the change in the concentration of the added L-Phe to the vesicles was presented in Figure 12b, where the inset of this figure justifies the decrement of the microviscosity with the reduced value of the τ_{slow}. Therefore, the freedom to the rotational restrictions of the probes inside the membrane bilayer is supposed to be associated with the reduced microviscosity of the membrane bilayer, the result of which is the increased fluidity or the decreased rigidity of the membrane.

3.3.4.3. Solvation Dynamics Study to Probe the Alteration of the Hydration Behavior of the Membrane in the Presence of L-Phe. Now, to understand whether the interaction of the L-Phe fibrils with the fatty acid membrane introduces an alteration of the hydration properties of the vesicle membrane or not, we performed a solvation dynamics experiment using the probe C-153. Since the vesicle membrane bilayer closely resembles a biological cell environment,[66] the consideration of the solvation dynamics of the associated water to this vesicle membrane is an important parameter to track. Here, the solvent correlation function $C(t)$, obtained from this experiment after fitting the time-resolved emission spectra (Figure S2, Supporting Information), is presented in Figure 13a in an exponential scale and Figure 13b in a logarithmic scale, whereas the corresponding extracted parameters after the biexponential fitting of the $C(t)$ function were depicted in Table 6. The justification for the utilization of the biexponential fitting model in the given context is associated with the fact that the aim is to explore the dynamics of two types of water associated with the membrane; one is "bound" water, and the other one is "exchange water". In the case of neat oleic acid vesicles, a time component of 3354 ps with the relative contribution of 16% was assigned to the bound water, whereas the component of 617 ps with the relative contribution of 84% was ascribed to the dynamics of the exchange water, which altogether estimates the average solvation time of 1054 ps. Interestingly, the addition of the L-Phe to the membrane at the highest concentration (12 mM) was found to introduce a significant alteration of both of these dynamic time scales of the bound as well as exchange water. Specifically, the dynamics of the exchange water time scales become 568 ps with the contribution of 88%, and the bound water shows the solvation time scale of 3150 ps with the relative contribution of 12%, which altogether provides the average solvation time scale of 877 ps. That means, the addition of the assemblies of the L-Phe to the vesicles increases the hydration dynamics of the water in and around the vesicle, which is a signature of the enhanced fluidity of the membrane. The alteration of the hydration dynamics time scales can be correlated with the alterations of rigidity or the fluidity of the bilayer membrane. For example, the reduced value of the solvation dynamics for the lipid vesicles in the presence of peptides has been correlated with the increased fluidity of the membrane.[62] Contrary to this, the sugar-induced elevation of the average solvation time of the lipid vesicles has been correlated with the enhanced rigidity of the membrane.[47] Therefore, in the present study, the conclusion regarding the elevated fluidity of the membrane reconciles the reflection obtained from the time-resolved anisotropy experiment, which also implies the greater rotational freedom of the probe molecules to rotate inside the membrane bilayer, thereby indicating a smaller extent of confinement in the presence of L-Phe. This free rotational freedom of the probes inside the membrane in the presence of L-Phe reconciles with the increased fluidity of the membrane in a sense of decreased rigidity.

Table 6. Picosecond Time-Scaled Solvent Correlation Function, $C(t)$ Decay Parameters of C-153 in Oleic Acid Vesicles in the Absence and Presence of L-Phe

system	τ_1(ps)	a_1	τ_2(ps)	a_2	τ_{avg}[a] (ps)
OA vesicles	617	0.84	3354	0.16	1054
OA vesicles + 12 mM L-Phe	568	0.88	3150	0.12	877

[a]Experimental error ±5%.

3.3.4.4. Cell Viability Study. Here, in order to evaluate the cell cytotoxicity of the L-Phe assemblies in the context of the cellular environment fed with the 20 mM oleic acid at pH 8.28, a cell viability study was performed. The details of the cell culture and the cell-viability assay protocol were discussed in the Supporting Information. Briefly, 3T3 mouse embryonic fibroblast cells fed with oleic acid were incubated with the L-Phe assemblies at various concentrations, and the 3-(4,5-dimethylthiazolyl-2)-2,5-diphenyltetrazolium bromide (MTT) assay was used for the quantification of the cell viability, as studied earlier.[9] The cell cytotoxicity was found to take place after the incubation of the media with the oleic acid, and further drops in the cell viability were observed when the oleic acid-containing cell media was treated with the L-Phe assemblies of various concentrations (Figure 14a). Specifically, the medium containing cells incubated with oleic acid was found to show a slight cytotoxicity, where the viability was ~90% (Figure 14a). Such an oleic acid-based cytotoxicity of a reduced order was also reported earlier on the other cellular environment, where an oleic acid-induced mitochondrial depolarization along with the lipid accumulation were reported to be the underlying factors for the oleic acid-induced cell death.[71] Additionally, an oleic acid-induced apoptosis of the tongue squamous cells is also recently reported, which certifies this compound to be a promising candidate for the selection of an anticancer compound against tumors.[72] However, when L-Phe was introduced to the cell media, the cell viability was found to drop considerably in a concentration-dependent manner (Figure 14a). Specifically, the viability drops to 70% and 65% in the presence of 1.2 and 2 mM L-Phe, respectively, while this value drops to 40% when 5 mM L-Phe was added (Figure 14a). On the one hand, this established the cytotoxic nature of the self-assembly formed from the various concentrations of the L-Phe. On the other hand, this concentration-dependent cytotoxicity is worth reconciling with the concentration-dependent effect of the L-Phe on the oleic acid vesicle membrane too, as discussed in the previous part of the manuscript. For example, the effect on the vesicle membrane was found to be much more prominent at the highest concentration of the L-Phe added, as observed from the time-resolved anisotropy measurement, solvation dynamics study, etc. On the one hand, likewise, the consequences at the higher concentration of the L-Phe are found to be much more significant in terms of the cytotoxic effect, as shown in Figure

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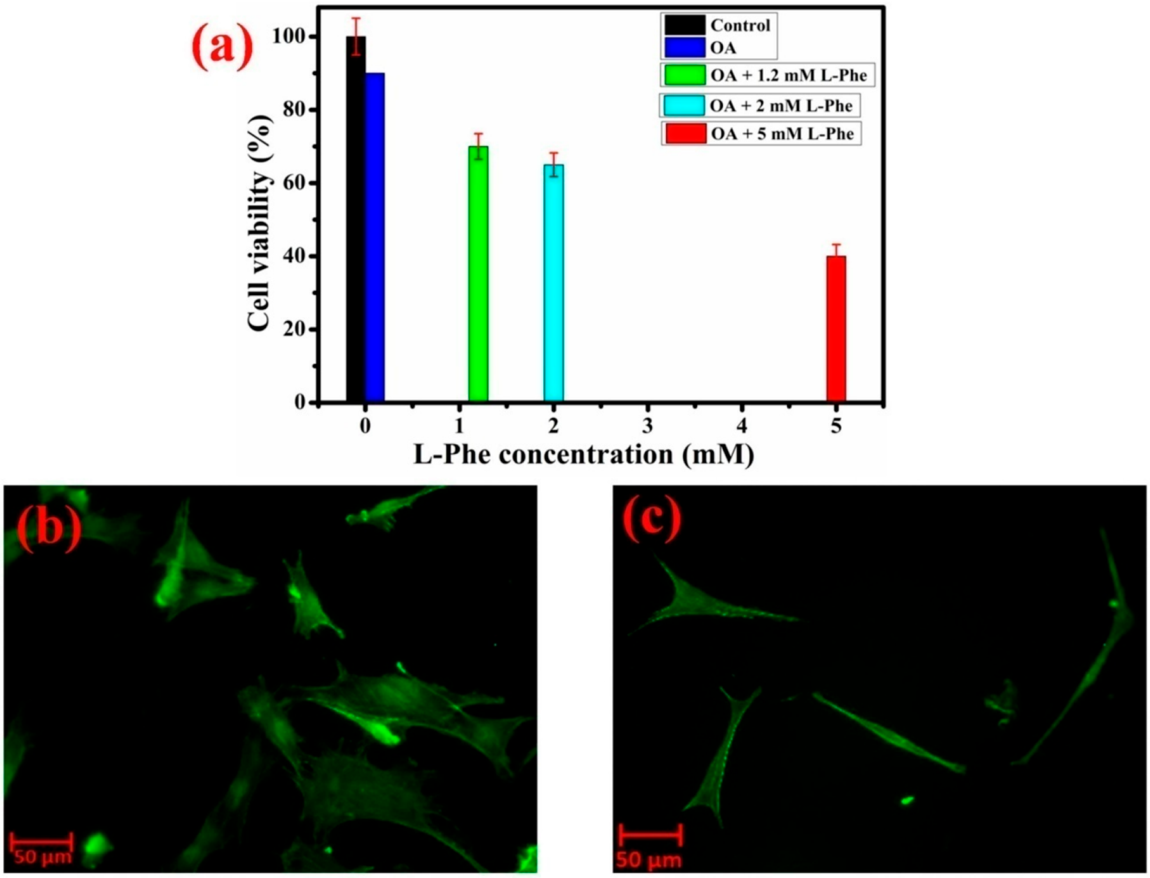

Figure 14. MTT assay-based cell viability study of the 3T3 mouse embryonic fibroblast cells incubated in the presence of OA and various concentrations of L-Phe (a). The error bars signify the alterations of the cell viability values after the triplicate mode of repetition of the experiment. Morphology of the cells incubated with OA in absence of L-Phe (b) and in the presence of 5 mM L-Phe (c).

14a. On the other hand, the alteration of the cell morphology is also found in the presence of L-Phe (Figure 14c) compared to the cells in the absence of L-Phe (Figure 14b). The self-assembled fibrillar morphologies of L-Phe are expected to impose damage on the cell membrane, the result of which is the change in the morphology of the cell (Figure 14c). Therefore, the cytotoxic effect of the assemblies of L-Phe on the cellular environment is expected to originate due to the imposition of the damaging effect of the assemblies on the cell membrane, as observed in the case of the nanoparticle cytotoxicity.[73]

4. CONCLUSIONS

The various supramolecular confined architectures formed by the self-assembly of oleic acid at various pH values have been characterized by a spectroscopic and microscopic investigation in this article. Oleic acid forms vesicles at pH ≈ 8.28, cylindrical micelles at pH ≈ 9.72, and spherical micelles at pH ≈ 11.77. The water dynamics in these confined media become faster on moving from vesicle to micelle. The anisotropy measurement confirms that the vesicle bilayer confinement at pH ≈ 8.28 is more restricted than the micellar confinement at pH ≈ 11.77, which in turn is less confined than at pH ≈ 9.72, where the cylindrical micelle is reported to be present mainly. The fluorescence correlation spectroscopy at a single molecular level also reveals that the self-assembled structure at pH ≈ 8.28 is more confined than the structures at higher pH values. Therefore, this dynamic formation of oleic acid vesicles originating from simple natural fatty acid micelles will strengthen the "origin-of-life" hypothesis in the sense that fatty acid micelles are more likely to form at the prebiotic age of the earth because of the small and simple structures, which alters over the age of the earth, thereby forming the complex compartmental structures of the first primitive cellular mimicking setup, that is, vesicles.

However, the self-assembled fibrillar morphologies of the essential amino acid L-phenylalanine, responsible for an in-born error of metabolic disorder like PKU, is found to strongly modulate the dynamics properties of the oleic acid vesicle membrane. Specifically, the interaction of the L-Phe assemblies with the oleic acid vesicle membrane reduces the rigidity of the vesicle membrane and alters the hydration properties of the membrane, as established from various spectroscopic and microscopic investigations. Such a fibril-induced alteration of the membrane properties is significant to be considered with due care in the biological context because of the fact that properties of the membrane, like rigidity, permeability, etc., can account for the change in the crucial biological processes, like the propagation of signals throughout the nerve cells and metabolic electron transport.[8] Hence the phenylalanine-induced alterations of the vesicle membrane properties are expected to be considered with due care for the extension of the therapeutic windows against a disorder like PKU, which arises due to the accumulation of fibrillar morphologies of L-Phe.

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■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jpcb.1c05592.

Instrumentation details, cell culture, and cell viability assay protocol, the values of translational diffusion coefficient (D_t) and the corresponding diffusion times (τ_D) of DCM and rhodamine-6G in various oleic acid assemblies at different pH values, steady-state fluorescence emission spectra of C-153 in L-Phe fibrillar solution, and time-resolved emission spectra (TRES) of C-153 in OA vesicle in absence and presence of L-Phe (PDF)

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Notes

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